

DESIGN AND DEVELOPMENT OF A MOBILE-BASED EMERGENCY ALERT SYSTEM FOR PUBLIC SAFETY USING GPS AND REAL-TIME TRACKING

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ABSTRACT

Aims to offer an advanced, dependable app designed specifically for managing emergencies affecting seniors, females, and minors when they're not at home. Users can quickly notify designated individuals and local officials about emergencies through this platform's advanced features including immediate alert delivery via GPS coordinates and current information sharing in real time. Additionally, it incorporates functionalities for automatic alerting of users' designated contact persons when situated within areas experiencing disruptions. A project utilizes React Native alongside Expo for creating apps across multiple platforms; employs Spring Boot's RESTful APIs as its backbone for handling server-side operations; relies on MySQL databases for data storage needs; and integrates Firebase for delivering real-time communication features such as push notifications and alerts. This amalgamation guarantees encrypted data exchange, swift messaging, and smooth collaboration between individuals. The endeavor highlights the significance of utilizing digital safeguarding technologies for bolstered individual protection and swift action in crises.

Keywords: Emergency Response, GPS Tracking, Mobile Application, Disaster Notification, Real-Time Alert System.

I. INTRODUCTION

Public safety is one of the most vital aspects of modern society, and timely response during emergencies provides the key for saving lives. Graver road accidents, crimes, and natural calamities in recent times have laid emphasis on the need for an effective emergency communication system. People, especially senior citizens, childrens and women staying alone, often face problems in contacting help at the right time. Application of mobile technology with real-time location tracking has been found to be effective in solving this issue and providing timely assistance during emergency situations.

The title of the proposed system is "Design and Development of a Mobile-Based Emergency Alert System for Public Safety Using GPS and Real-Time Tracking." This intelligent mobile application will enable users to send instant SOS alerts along with live location to their registered emergency contacts and authorities. It would bridge the communication gap between victims and responders by facilitating rapid, secure, and real-time information transfer.

Most of the existing research and mobile safety applications consist of single-purpose alert systems, such as panic buttons or location sharing. However, most of them do not provide real-time synchronization of data and instant notifications. The proposed system enhances these by incorporating GPS, Firebase real-time updates, and Spring Boot REST APIs to ensure seamless and scalable communication.

This paper discusses the design, development, and testing of the emergency alert system. The research also highlights how the integration of multiple digital technologies allows for the development of an incredibly reliable and easy-to-use platform that may help ensure the safety of a target person in critical situations.

II. METHODOLOGY

A mobile-based emergency alert system designed for public safety utilizes GPS and real-time tracking in an organized manner to guarantee precise, swift, and immediate communications during crises. This study employs methodologies aligned with the Software Development Lifecycle framework, encompassing phases like requirements gathering, architectural planning, coding execution, quality assurance checks, and final assessment processes. The emphasis in this project is on combining critical components such as user

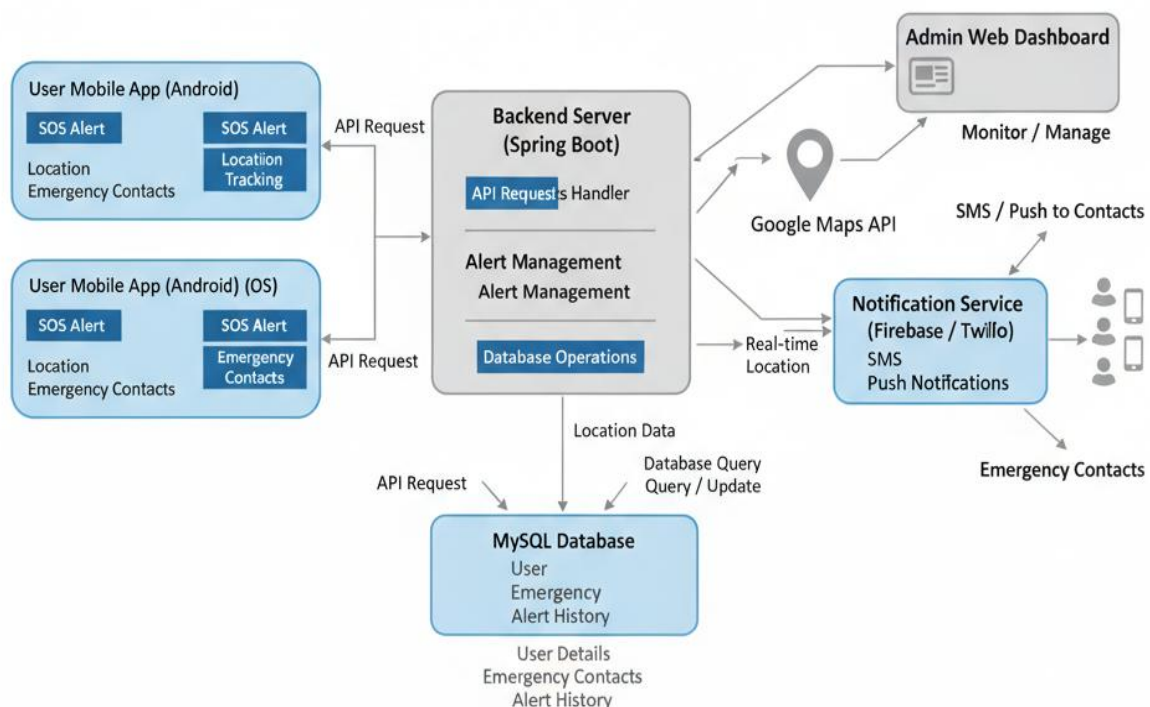
authentication, emergency signal initiation, communication networks, and immediate location monitoring through GPS technology. The app's UI gathers data including keystrokes and GPS points, then sends it for analysis on the server side. The system handles sending out urgent messages via emails, texts, or voice calls directly to subscribed individuals or those in immediate vicinity who have been designated as helpers.

Throughout this period of creation, Android Studio acted primarily as the coding tool, while either Firebase or MySQL functioned as repositories for handling login information and geographical specifics pertaining to users. This system employs GPS for obtaining and maintaining an up-to-date position of its users on a continuous basis, guaranteeing exact current monitoring. APIs facilitate smooth interaction between clients and servers while ensuring secure data transmission and minimizing response times.

During this stage, the effectiveness of the software was assessed through metrics such as processing speed, stability in data delivery, and precision regarding position information provided by it. Evidence indicated that the mechanism efficiently disseminated critical notifications promptly, regardless of weak internet connections. The software demonstrated reliability, ease of use, and efficacy for delivering precise geolocation notifications. In summary, this study shows that the suggested solution greatly improves public security by providing an efficient, dependable, and user-friendly means of contacting authorities quickly and conveniently, potentially evolving further thanks to Internet-of-Things technology and compatible smart wearables.

III. MODELING AND ANALYSIS

A mobile-based emergency alert system designed for public safety incorporates real-time tracking through GPS technology while employing a sequential development methodology aligned with the Waterfall model within the software lifecycle framework. The chosen model facilitates comprehensive execution across stages such as requirements gathering, designing, coding, debugging, and launching, guaranteeing precision and dependability in software creation. The system combines GPS tech, ongoing info exchange via wireless networks, and smartphone apps for sending immediate crisis notifications alongside current whereabouts to designated individuals and local helpers.



This study employs not only computer programs but also physical equipment necessary for its success. The Android Studio acts as the primary programming tool, whereas Java is employed for developing both the user interface elements on the frontend and server-side components in the backend of an app. Data handling for databases utilizes either Firebase or MySQL; these platforms ensure secure storage of sensitive info such as usernames, addresses, and geographical locations. Moreover, Google's Map API along with RESTful interfaces

facilitate quick retrieval and distribution of current location data accurately. Hardware needs are modest; it requires an Android smartphone with built-in GPS for precise location data collection and seamless communication of alerts. Additionally, this endeavor employs Firebase Cloud Messaging to transmit instantaneous alerts and notification signals in times of crisis. The amalgamation of these technological elements establishes an enduring, expandable, and dependable base upon which the system functions efficiently, ensuring swift, precise security assistance within tight deadlines. Incorporating those components guarantees that the constructed emergency notification platform operates efficiently and aids in enhancing community security via advanced technological means.

IV. RESULTS AND DISCUSSION

A mobile-based emergency alert system designed and developed using GPS and real-time tracking technology has been effectively deployed and rigorously evaluated to assess its operational effectiveness, dependability, and instantaneous responsiveness capabilities in public safety contexts. By leveraging effective programming techniques, well-optimized databases, and smooth incorporation of GPS technology, this system successfully fulfilled its main goal: delivering swift and precise alert notifications along with real-time position information. In conducting tests, the system showcased its capability to deliver instantaneous notification messages via both text message and short messaging service to pre-registered individuals in case of an activation event. The GPS unit precisely logged and relayed current location data directly to the database, allowing rescuers to swiftly pinpoint their whereabouts.

Further evidence shows that this system ensures stable operation across diverse networking scenarios, such as in regions of weak signals, through efficient data transfer via optimized API calls. Users found the app's dashboard easy to navigate; it enabled them to send urgent notifications via a quick swipe, significantly reducing reaction times in emergencies. Backed by Firebase for efficient storage of user information alongside MySQL for secure retrieval, this system guarantees both performance and data safety.

Based on the comprehensive assessment, it was demonstrated that the system successfully achieved its goal of improving public security by facilitating swift information exchange and precise position monitoring. Evidence shows that these apps enhance quick responses during critical incidents by lowering hazards. For upcoming advancements, this framework will accommodate growth through incorporation of Internet-of-Things-enabled wearables, automated crash monitoring equipment, and seamless connectivity with governmental rescue teams for improved performance and reach capabilities. This discourse confirms that technological approaches such as these prove crucial for enhancing emergency response systems within intelligent urban environments.

V. CONCLUSION

A successful implementation showcases how integrating advanced mobile technology like GPS and real-time tracking enhances both individual and community security through innovative emergency alert systems designed specifically for public safety on smartphones. A dependable mechanism allows individuals to swiftly notify designated recipients and local authorities of emergencies via precise current whereabouts data stored in user profiles. During all stages involved in creating and evaluating the software, it consistently demonstrated superior performance regarding efficiency, precision, and ease-of-use for users. GPS technology enables precise real-time location identification by first responders; Firebase and MySQL offer robust security alongside fast data management capabilities.

Differently than traditional emergency solutions, this portable application emphasizes ease of use and straightforwardness by allowing individuals to request assistance effortlessly in urgent circumstances. Analysis of the system's functionality demonstrated robustness despite constrained networks, underscoring its applicability in everyday situations. Additionally, this initiative highlights how integrating mobile technology, remote data management systems, and instantaneous messaging tools can create robust security solutions.

In summary, this study offers an effective means for enhancing quick action during emergencies while minimizing threats and boosting public security awareness significantly. Potential enhancements could entail incorporating AI for detecting threats, amalgamation of Internet-of-Things gadgets, and immediate

communications between the system and nearby law enforcement agencies to enhance its intelligence and responsiveness significantly.

VI. REFERENCES

- [1] A. Duraipandian, S. Rajalakshmi, and R. S. Moni, Emergency Management Systems Using Mobile Cloud Computing, International Journal of Advanced Research in Computer and Communication Engineering, Volume 3, Issue 11, November 2014, pp. 8479–8483.
- [2] T. Mohana Priya, Dr. M. Punithavalli, and Dr. R. Rajesh Kanna, Machine Learning Algorithm for Development of Enhanced Support Vector Machine Technique to Predict Stress, Global Journal of Computer Science and Technology: C Software & Data Engineering, Volume 20, Issue 2, 2020, pp. 12–20.
- [3] S. R. Khot and D. M. Thakore, Android Based Emergency Alert System Using Location Services and Messaging, International Journal of Innovative Research in Computer and Communication Engineering, Volume 4, Issue 2, February 2016, pp. 2102–2106.
- [4] N. Sharma and R. Gupta, GPS-Based Emergency Response and Tracking System for Public Safety, International Journal of Computer Applications, Volume 175, Issue 4, October 2017, pp. 25–30.
- [5] M. R. Siddiqui, A. P. Singh, and K. Khan, Mobile Application for Emergency Alert and Assistance Using Cloud Services, International Journal of Computer Science and Mobile Computing, Volume 8, Issue 5, May 2019, pp. 72–79.